

Yash Kherwal

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EDUCATION

Indian Institute of Technology Roorkee

2022 – 2026

B.Tech in Chemical Engineering

EXPERIENCE

Product Manager Intern

Feb 2026 – Apr 2026

OrangeLabs (EdTech)

- Contributed to **0-to-1 product development** of an AI-powered school ERP — defined product vision, user personas, and feature scope for two flagship AI products from initial concept to structured requirements.
- Shaped **AI Attendance System**: identified teacher time-on-task as the primary pain point; scoped CCTV-based auto-detection as the solution, eliminating register-based roll calls and freeing ~10 minutes per class for instruction.

Software Development Intern

May 2025 – June 2025

ScoreMe Solutions

- Scoped and shipped a **two-stage PDF classification pipeline** for fintech document processing — rule-based fast path for deterministic inputs, **Random Forest fallback** for ambiguous cases; **automated ~80% of document volume**, reducing analyst review queue to exception-only cases.
- Defined **confidence threshold** as an explicit **human-in-the-loop gate**: outputs withheld from **credit scoring pipelines** and flagged for review; extended coverage to scanned PDFs via **Tesseract OCR**, doubling addressable input types.

Engineering Intern

June 2025 – July 2025

Bharat Electronics Limited

- Built a real-time stream processor for **ASTERIX CAT048** radar protocol per **EUROCONTROL** spec; decoded variable-length packet structures into structured aviation records; designed **concurrent ingestion** with **backpressure handling** to sustain throughput under high-frequency surveillance loads.

PROJECTS

AI Data Analyst Agent | *Python, LangGraph, CrewAI, n8n, MCP, FastAPI, Docker, AWS EC2*

2025

- Built a **5-agent LangGraph workflow** (Query Planner → Code Writer → Executor → Error Fixer → Insight Generator) — **reduced time-to-insight from hours to under 60s**; Query Planner routes deterministically to **SQL or Python** by source type; supports **CSV, SQL databases, and PDF** inputs.
- **Error Fixer agent** classifies exceptions, regenerates fix, retries up to **3 attempts** before surfacing to user; **n8n** orchestrates triggers; **MCP** exposes files, DBs, and APIs as tool context; Executor in **Docker sandbox**; **FastAPI** on **AWS EC2**.

Semantic Document Search — **GDSC IIT Roorkee** | *Python, LangChain, BGE-M3, AstraDB, FastAPI, AWS EC2*

2024

- Replaced keyword search on a **500+ document corpus** with **hybrid RAG** (dense **BGE-M3** + sparse BM25) — eliminated zero-result failure mode; users receive grounded direct answers instead of document lists.
- Offline embedding pipeline into **AstraDB** decoupled from **FastAPI** serving on **AWS EC2** — **sub-second query response**; corpus updates without downtime; prompt constraints enforce **response groundedness**.

YAAR — AI Behavioral Companion | *React Native, LLM APIs, FastAPI, EAS Build*

2025

- **Shipped Android APK** (React Native + EAS Build) — tap-based reaction probes, adaptive LLM generation, shareable identity card export; **sub-2s streamed response latency**; validated end-to-end on device.
- Preference engine tracks **3 personality dimensions** (humour tolerance, self-awareness index, reaction intensity) — steers LLM tone via dynamic prompt construction; **personalised output from first session** deepening across interactions without retraining.
- Designed **zero-input engagement loop** (Open → Hook → React → Preference Update) for Gen Z in Tier 2/3 India; revenue via **contextual native brand integration** triggered by behavioral state at higher CPM than display.

Echo Pod — AI Stillness Companion | *Android, On-Device Voice AI, Event-Driven Triggers*

2025

- Diagnosed **25% first-session churn** on screen blocker apps; designed a **3-layer AI UX** (ambient trigger → on-device voice AI → daily stillness score) with **on-device inference** for zero latency and privacy by default; **pause completion rate** as primary activation signal.

TECHNICAL SKILLS

AI/ML: LangGraph, LangChain, CrewAI, MCP, Hybrid RAG, Multi-Agent Orchestration, Prompt Engineering, Human-in-the-Loop Design, Retrieval Evaluation

Product: PRD Writing, MVP Scoping, Prioritisation (RICE, Knapsack), A/B Testing, Funnel Analysis, Latency–Accuracy Tradeoffs, Failure Handling

Backend & Infra: Python, FastAPI, Docker, AWS EC2, n8n, Stream Processing, REST APIs

Data & Analytics: SQL (window functions, cohort queries), Pandas, OOP, DSA (graphs, trees, complexity)